

KRISHNA Koushik Thokala

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Summary

Cloud Infrastructure Engineer and MS CS graduate with 2+ years in production NOC operations, incident response, and SLA-driven troubleshooting across Tier-1 carrier networks. Designs and builds AWS platform projects across EKS, Terraform, GitOps, serverless, CI/CD, observability, and SRE workflows.

Technical Skills

Cloud & AWS: AWS, VPC, IAM, EC2, S3, Lambda, API Gateway, EventBridge, CloudWatch, SNS, DynamoDB, CloudFront, EKS, ECR, RDS, ALB, Secrets Manager, Bedrock
Infrastructure & DevOps: Terraform, Docker, Kubernetes, Helm, Argo CD, GitHub Actions, GitOps, Git, CI/CD, Linux
Observability & SRE: CloudWatch Logs/Metrics/Alarms, k6 Load Testing, HPA Autoscaling, Incident Response, RCA, Runbooks, SLA Troubleshooting, MTTR, SLO Metrics
Networking: TCP/IP, DNS, BGP, OSPF, MPLS, VPN, Load Balancing, Firewalls, Network Security, DWDM, Juniper Routing
Programming & Databases: Python, Bash, SQL, PostgreSQL, REST APIs, FastAPI, React
Certifications: AWS Certified Cloud Practitioner (2026)

Experience

Tata Communications

July 2022 – July 2024

Senior Engineer, Shift Lead (production NOC operations)

Pune, India

- Owned high-severity incidents across 25-30 Tier-1 carrier clients on global ILL, DWDM, and submarine cable networks, driving triage, TAC/vendor coordination, field dispatch, and restoration.
- Diagnosed and resolved BGP/MPLS failures, DWDM optical faults, and equipment malfunctions across Juniper, Huawei, Ciena, and Alcatel platforms using segment-wise isolation, restoring service within SLA on carrier-grade global networks.
- Led 5-person global NOC shifts through 40+ daily priority incidents, including AVP/VP-level escalations, coordinating handoffs, RCA notes, customer-premises testing, and cross-functional follow-up until resolution.
- Restored high-priority services within 4-hour SLA targets and contributed to 99.9% annual network availability through TAC, field, and vendor escalations; earned Certificate of Excellence.

Projects

CloudOps SRE Platform | EKS, Terraform, Helm, FastAPI, React, RDS

May 2026 – June 2026

- Built an EKS-based SRE operations platform centralizing service health, deployment history, incident timelines, MTTR metrics, SLO tracking, and error budget views for cloud operations workflows.
- Deployed FastAPI and React workloads behind ALB Ingress on EKS with Helm-managed Kubernetes deployments, services, resource limits, liveness/readiness probes, and Secrets Manager-backed database credentials.
- Validated autoscaling under k6 load as backend replicas scaled from 2 to 6 pods, sustaining 2,035 requests at a 99.5% success rate with 1.52s p95 latency; confirmed behavior through HPA status and CloudWatch logs.

CloudOps GitOps Platform | EKS, Argo CD, Helm, Terraform, GitHub Actions

June 2026

- Built a GitOps delivery platform on EKS using Argo CD, Helm, and GitHub Actions, with multi-source Applications and environment-specific Helm values for namespace-isolated dev/staging/prod environments.
- Provisioned VPC/EKS/ECR/IAM with Terraform; added ResourceQuotas, scoped RBAC, and PR-style image promotion through Git-managed Helm values; validated drift self-healing and Git revert rollback on EKS using a Helm-toggled `failureMode` readiness failure.

CloudOps Uptime Monitor | Lambda, DynamoDB, API Gateway, Terraform

April 2026 – June 2026

- Built serverless uptime monitoring platform that checks website availability every 5 minutes using EventBridge, Lambda, DynamoDB, API Gateway, SNS, CloudWatch, Terraform, and React on S3/CloudFront.
- Optimized current-status reads from 86,400 history records to 10 latest-status rows for a 10-URL, 30-day workload; added API keys, throttling, restricted CORS, KMS, DLQs, state-change alerts, 9 CloudWatch metrics, and CI quality gates.

Education

Indiana University Bloomington

August 2024 – May 2026

Master of Science in Computer Science, GPA: 3.5/4.0

Bloomington, IN

SRM Institute of Science and Technology

June 2018 – June 2022

Bachelor of Technology in Computer Science and Engineering, CGPA: 8.43/10

Chennai, India